

SNEHA RATHI

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Result-oriented Software Engineer with 2+ years of experience specializing in full-stack development, cloud-native solutions, and machine learning. Proficient in Python, Java, AWS (EC2, S3, Docker, Kubernetes), and Flask/FastAPI, with a track record of deploying scalable ML models and CI/CD pipelines. Passionate about optimizing performance and building robust applications with modern tech stacks. Strong problem-solver with expertise in REST APIs, data pipelines, and real-time systems.

TECHNICAL SKILLS

Programming Languages: C, Java, Python, SQL, JavaScript

Web and Cloud Technologies: HTML, CSS, Flask, FastAPI, PHP, AWS (EC2, S3, Docker, Kubernetes), Google Maps API

Developer Tools: Git, GitHub, Jenkins, Eclipse, Rancher, PyTest, GitHub Actions, Pickle

Data and ML Tools: Pandas, GridSearchCV, SVM, LeetCode, HackerRank

Operating Systems and Embedded: Windows 11, MS-DOS, POSIX, Arduino

PROFESSIONAL EXPERIENCE

Intellify **June 2023**

Software Engineering Teaching Intern

- Led 10+ workshops on OOP, data structures, and algorithms, improving students' coding proficiency in Java.
- Mentored students in live code reviews and pair programming, improving their debugging and problem-solving abilities.
- Integrated Git and GitHub into lessons, training students in collaborative coding and essential version control practices.
- Utilized HackerRank and LeetCode to assess student code, offering instant feedback and personalized improvement plans.
- Guided new students in IDE setup and dev environment configuration for smooth onboarding and technical readiness.

SmartKrawler **May 2021**

AI/ML Software Engineer Intern

- Built SVM image classifier to distinguish apple vs. mango images, addressing data imbalance, ensuring robust deployment.
- Developed REST APIs using Flask/FastAPI for real-time model serving, enabling easy integration with client applications.
- Created ETL data pipelines in Python (Pandas) for scalable data ingestion, transformation, and training dataset preparation.
- Utilized GridSearchCV for hyperparameter tuning, resolving underfitting/overfitting, optimizing SVM model performance.
- Evaluated model through confusion matrix, precision, recall, and F1-score to ensure reliability and address bias/variance.
- Serialized model with Pickle for real-time prediction and seamless deployment.
- Secured ML workflows by anonymizing sensitive data and enforcing access controls as per GDPR/CCPA requirements.

PROJECTS

CI/CD Pipeline and Containerized Web App Deployment **Mar 2025**

- Containerized a Java web app through Docker, orchestrated deployment on Kubernetes (Rancher) with 3 pods for HA.
- Built Jenkins CI/CD with GitHub, authored Dockerfile/Jenkinsfile/YAML, and deployed on AWS EC2 for public access.

Kitchen Alchemy – Recipe Management Web App **Dec 2024**

- Built a recipe management web app with search, review, favorite features, boosting user meal planning efficiency.
- Enhanced UI/UX using filtering, category sorting, and responsive navigation for seamless user experience.

Automatic Fire Response System (Real-Time Operating System Project) **Dec 2024**

- Engineered a real-time fire alert system using DHT22, GPIO, and buzzer, prioritizing sensor reads with POSIX threads.
- Developed console interface for live temperature, humidity, fire status, ensuring real-time system responsiveness.

Movie Database Management System **Feb 2024**

- Modeled data through EER diagrams, implemented a relational Oracle SQL database for secure, efficient movie storage.
- Wrote advanced SQL queries for fast retrieval of movie information, reviews, awards, and distributor details.

Baggage Tracking System using IoT **Jan 2023**

- Developed IoT-based baggage tracker using Arduino, Neo6m GPS, and SIM800L GSM, enabling live location monitoring.
- Created a PHP web interface with Google Maps API to display and store baggage location for security in transit.

Accident Alert and Road Condition Reporting App **Aug 2022**

- Designed a web/mobile app with geotagged complaint submission, optional images, and dynamic officer assignment.
- Integrated real-time complaint status tracking with display by National Highway number for user transparency.

EDUCATION

George Mason University, Fairfax, VA | Master of Science; Major in Computer Science **Aug 2023 – May 2025**

Keshav Memorial Institute of Technology | Bachelor of Technology; Major in Computer Science **Aug 2019 – May 2023**

ADDITIONAL EXPERIENCE

- Led cross-functional kitchen operations as a Kitchen Leader with ServSafe Certification at Chipotle, demonstrating strong problem-solving and time management skills in high-pressure environments.
- Maintained composure and made logical decisions under stress while effectively communicating with team members.